
Upper-Linearizability of Online Non-Monotone DR-Submodular Maximization over Down-Closed Convex Sets

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Abstract

We study online maximization of non-monotone Diminishing-Return(DR)-submodular functions over down-closed convex sets, a regime where existing projection-free online methods suffer from suboptimal regret and limited feedback guarantees. Our main contribution is a new structural result showing that this class is $1/e$ -linearizable under carefully designed exponential reparametrization, scaling parameter, and surrogate potential, enabling a reduction to online linear optimization. This allows us to obtain first non-Frank-Wolfe type algorithms for this setting that obtain an approximation coefficient better than $1/4$. Moreover, the linearization framework allows us to move beyond offline optimization. As a result, we obtain $O(T^{1/2})$ static regret with a single gradient query per round and unlock adaptive and dynamic regret guarantees, together with improved rates under semi-bandit, bandit, and zeroth-order feedback. Across all feedback models, our bounds strictly improve the state of the art.

1. Introduction

Online optimization of submodular and DR-submodular functions has become a central primitive in machine learning, with applications in mean-field inference, revenue maximization, influence maximization, supply chain management, power network reconfiguration, and experimental design (Bian et al., 2019; Ito & Fujimaki, 2016; Gu et al., 2023; Aldrichetti et al., 2021; Mishra et al., 2017; Li et al., 2023). In these problems, an algorithm repeatedly selects actions from a convex domain while an adversary

reveals a reward function, and performance is measured through notions of static, adaptive, or dynamic regret. A long line of work has developed projection-free algorithms, typically based on Frank–Wolfe or boosting-style updates (Fazel & Sadeghi, 2023; Chen et al., 2018; Zhang et al., 2022; Pedramfar et al., 2023; Zhang et al., 2024).

While most of the study in DR-submodular optimization is for monotone objectives (Hassani et al., 2017; Fazel & Sadeghi, 2023; Chen et al., 2018; Zhang et al., 2022), the non-monotone objective plays an important role in many applications such as price optimization, social networks recommendation, and budget allocation (Ito & Fujimaki, 2016; Gu et al., 2023; Alon et al., 2012). In this work, we study non-monotone DR-submodular maximization over down-closed convex sets, which remains particularly challenging. Down-closed domains include box constraints, knapsack polytopes, and intersections of matroids, and arise naturally in resource allocation and coverage problems. In this regime, even the best approximation ratio for online optimization remains open (Buchbinder & Feldman, 2024), while the best achievable approximation rate is $1/e$ (Thang & Srivastav, 2021; Zhang et al., 2023; Pedramfar et al., 2023). Further, for this regime, existing projection-free methods either require multiple oracle queries per round or only achieve suboptimal regret rates such as $O(T^{2/3})$ (Pedramfar et al., 2024a), and essentially no results were known for adaptive or dynamic regret.

Recent work in Pedramfar & Aggarwal (2024b) introduced the notion of *linearizable* function classes and showed how such a structure enables a generic reduction from online linear optimization to a wide family of non-convex problems, including several DR-submodular settings. While powerful, these results does not cover the non-monotone down-closed case with efficient query complexity and $O(T^{1/2})$ regret. In this paper, we close this gap by establishing a new structural characterization for online non-monotone DR-submodular maximization over down-closed convex sets. Our main technical contribution is to show that this class is $1/e$ -linearizable under a carefully designed exponential reparametrization and surrogate potential.

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Technical Novelty. Achieving this result requires overcoming two fundamental theoretical barriers. First, establishing the structural $1/e$ -linearizability reduction (Theorem 1) requires carefully managing the non-monotone penalty over down-closed sets. We achieve this by designing a novel surrogate potential $F(\cdot)$ that acts as an integrating factor, heavily weighting the early stages of the trajectory to enable an exact mathematical cancellation during the integration-by-parts analysis. Second, to translate this structure into an efficient online algorithm, we bypass the heavy computational cost of standard continuous greedy approximations by introducing a Jacobian-corrected gradient estimator (BQND, Algorithm 1). Unlike naive sampling, this estimator explicitly incorporates the curvature of our exponential mapping ($h(x) = 1 - e^{-x}$) into the gradient query, constructing unbiased linear surrogates of the non-monotone objective using a strict single-query budget. Together, these innovations completely decouple the non-convex oracle queries from the constraint handling, allowing our framework to accept any efficient regret-minimizing algorithm for linear functions as a base learner.

Once this structure is in place, a broad collection of algorithmic guarantees follow as immediate consequences through existing regret-transfer principles in (Pedramfar & Aggarwal, 2024b). We obtain the first projection-free online algorithms achieving $O(T^{1/2})$ static regret with only a single gradient query per round. Moreover, the same framework unlocks adaptive and dynamic regret guarantees in adversarial environments, as well as improved rates under semi-bandit, bandit, and zeroth-order full-information feedback. Across all feedback models, our results strictly improve the previously best-known bounds, as summarized in Table 1.

Summary of Contributions. Our primary contribution is breaking the theoretical bottleneck that historically trapped $1/e$ -approximation algorithms in $O(T^{2/3})$ regret. Specifically:

1. **Structural Breakthrough (Theorem 1):** We prove that non-monotone DR-submodular functions over down-closed convex sets are $1/e$ -linearizable. This result mathematically decouples the best-known online approximation ratio from Frank-Wolfe mechanics, enabling a clean reduction to Online Linear Optimization (OLO).
2. **Algorithmic Enabler (BQND, Algorithm 1):** We introduce a novel Jacobian-corrected gradient estimator that constructs unbiased linear surrogates of the non-monotone objective using a strict single-query budget, bypassing the computational cost of standard continuous greedy approximations.
3. **Best-known & Non-Stationary Regret Guarantees (Table 1):** By seamlessly plugging our structural theorem

into existing OLO meta-algorithms, a broad suite of state-of-the-art guarantees naturally follows as direct corollaries. We obtain best-known $O(T^{1/2})$ static regret with $O(1)$ queries, and unlock the first adaptive and dynamic regret guarantees for this setting, alongside improved rates across limited feedback settings.

2. Related Works

Online Non-monotone DR-Submodular Maximization. While the online maximization of monotone DR-submodular functions (Zhang et al., 2022; Hassani et al., 2017; Chen et al., 2018; Fazel & Sadeghi, 2023; Pedramfar et al., 2023; 2025; Lu et al., 2025) is well-understood, permitting efficient greedy solutions (Streeter & Golovin, 2008), the non-monotone regime presents significantly greater challenges. Algorithms must balance standard greedy ascent steps with corrective reduction steps to navigate the non-monotone landscape (Bian et al., 2017). Recently, Pedramfar & Aggarwal (2024b) proposed the elegant framework of *linearizable* functions, which reduces complex non-convex optimization problems, including monotone DR-submodular maximization (Pedramfar & Aggarwal, 2024b), non-monotone DR-submodular maximization under general convex set (Pedramfar & Aggarwal, 2024b), regularized phase retrieval problem (Sarkar et al., 2025), one-sided smooth function optimization (Pedramfar & Aggarwal, 2026), to Online Linear Optimization (OLO). As a special case, these approaches yield $1/4$ approximation ratio for non-monotone DR-submodular functions under general convex constraint set containing the origin, which is the optimal approximation coefficient for this class of problems. (See Mualem & Feldman (2023)).

The Down-Closed Convex Sets & The Frank-Wolfe Bottleneck. Overcoming this $1/4$ limit requires focusing on specific constraint geometries, such as down-closed convex sets. Even in the offline setting, finding the optimal approximation ratio over down-closed sets has remained an open challenge for over a decade. While an absolute upper bound of 0.478 exists (Gharan & Vondrák, 2011; Qi, 2024), the best achievable offline rate is currently 0.401 (Buchbinder & Feldman, 2024). In the online setting, the best known achievable approximation is $1/e$ (Thang & Srivastav, 2021; Zhang et al., 2023). Historically, achieving this $1/e$ ratio online has been strictly bottlenecked by a reliance on Frank-Wolfe (FW) and continuous greedy algorithms. This reliance introduces a severe trade-off: FW mechanics couple the non-convexity to the update rules, trapping the regret at $O(T^{2/3})$ for single-query algorithms (Pedramfar et al., 2024a) or requiring computationally expensive batch queries of $O(T^{3/2})$ to achieve $O(T^{1/2})$ regret (Zhang et al., 2023). This Frank-Wolfe

Table 1. Comparison of Regret Bounds for Online Non-Monotone DR-Submodular Maximization over Down-Closed Sets. **Oracle** denotes the feedback type (∇F : Gradient, F : Value). Our framework is the **first** to provide Adaptive and Dynamic regret guarantees while achieving $\mathcal{O}(T^{1/2})$ static regret in the $\mathcal{O}(1)$ query regime. Thang & Srivastav (2021) achieves $\mathcal{O}(T^{3/4})$ regret with $T^{3/4}$ queries per round when $\beta = \frac{3}{4}$. Zhang et al. (2023) achieves $\mathcal{O}(T^{1/2})$ regret with $T^{3/2}$ queries per round when $\beta = \frac{3}{2}$, but only achieves $\mathcal{O}(T^{4/5})$ regret with $\mathcal{O}(1)$ queries.

ORACLE	FEEDBACK	REFERENCE	APPROX.	QUERIES	REGRET GUARANTEES		
					STATIC	ADAPTIVE	DYNAMIC ($\times \sqrt{1 + P_T}$)
∇F	FULL INFO	(THANG & SRIVASTAV, 2021)	$1/e$	$T^\beta, \beta \in [0, \frac{3}{4}]$	$\mathcal{O}(T^{1-\beta/3})$	—	—
		(ZHANG ET AL., 2023)	$1/e$	$T^\beta, \beta \in [0, \frac{3}{2}]$	$\mathcal{O}(T^{1-\beta/3})$	—	—
		(ZHANG ET AL., 2023)	$1/e$	1	$\mathcal{O}(T^{4/5})$	—	—
		(PEDRAMFAR ET AL., 2024A)	$1/e$	1	$\mathcal{O}(T^{2/3})$	—	—
		THIS PAPER	$1/e$	1	$\mathcal{O}(T^{1/2})$ (PROP. 1)	$\mathcal{O}(T^{1/2})$ (PROP. 2)	$\tilde{\mathcal{O}}(T^{1/2})$ (PROP. 3)
	SEMI-BANDIT	(ZHANG ET AL., 2023)	$1/e$	1	$\mathcal{O}(T^{4/5})$	—	—
		(PEDRAMFAR ET AL., 2024A)	$1/e$	1	$\mathcal{O}(T^{3/4})$	—	—
		THIS PAPER	$1/e$	1	$\mathcal{O}(T^{2/3})$ (PROP. 4)	$\mathcal{O}(T^{2/3})$ (PROP. 4)	$\tilde{\mathcal{O}}(T^{2/3})$ (PROP. 7)
	FULL INFO	(PEDRAMFAR ET AL., 2024A)	$1/e$	1	$\mathcal{O}(T^{4/5})$	—	—
		THIS PAPER	$1/e$	1	$\mathcal{O}(T^{3/4})$ (PROP. 5)	$\mathcal{O}(T^{3/4})$ (PROP. 5)	$\tilde{\mathcal{O}}(T^{3/4})$ (PROP. 7)
F	BANDIT	(ZHANG ET AL., 2023) (DET.)*	$1/e$	1	$\mathcal{O}(T^{8/9})$	—	—
		(PEDRAMFAR ET AL., 2024A)	$1/e$	1	$\mathcal{O}(T^{5/6})$	—	—
	THIS PAPER	THIS PAPER	$1/e$	1	$\mathcal{O}(T^{4/5})$ (PROP. 6)	$\mathcal{O}(T^{4/5})$ (PROP. 6)	$\tilde{\mathcal{O}}(T^{4/5})$ (PROP. 7)

*Without indication of deterministic (Det.), the oracle is by default assumed to be stochastic with noise.

monopoly prevents the use of off-the-shelf OLO/OCO algorithms, and our work breaks this exact bottleneck by proving that $1/e$ can be achieved while preserving a clean linearizable reduction to OCO. Our algorithm is the first to simultaneously achieve $\mathcal{O}(T^{1/2})$ $1/e$ -regret and $\mathcal{O}(1)$ query efficiency in the down-closed regime.

Non-Stationary and Limited Feedback While dynamic (Zinkevich, 2003; Zhang et al., 2018; Zhao et al., 2021) and adaptive (Hazan & Seshadhri, 2009; Garber & Kretzu, 2022) regret bounds are well-established for convex optimization, these guarantees remain less explored for non-monotone DR-submodular functions. The combination of non-convexity and environmental drift presents a formidable barrier that standard expert-tracking meta-algorithms fail to address. Recent unified projection-free frameworks developed by Pedramfar et al. (2024a) have successfully addressed various feedback models (semi-bandit, bandit) for adversarial DR-submodular optimization, but they stop short of providing adaptive and dynamic regret guarantees for the challenging non-monotone, down-closed setting. We provide the first dynamic regret guarantees ($\tilde{\mathcal{O}}(\sqrt{T(1 + P_T)})$) for this problem class. Furthermore, we extend these robust guarantees to semi-bandit and bandit feedback settings, offering better results for non-stationary maximization under limited information of non-monotone DR-submodular functions over down-closed convex set.

3. Preliminaries

3.1. Notations

We use $\mathcal{F}$ to denote the function class to which all objective functions f_t belong. We use $\mathcal{A}$ to denote an online optimization algorithm. We denote vectors as boldface lower-case letters (e.g., $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$) and denote their coordinates as x_i, y_i . For any two vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we denote their element-wise product by $\mathbf{x} \odot \mathbf{y}$ and their element-wise exponential by $e^{\mathbf{x}}$. The inequality $\mathbf{x} \leq \mathbf{y}$ is understood coordinate-wise. We define the coordinate-wise probabilistic sum as $\mathbf{x} \oplus \mathbf{y} \triangleq \mathbf{1} - (\mathbf{1} - \mathbf{x}) \odot (\mathbf{1} - \mathbf{y})$.

Constraint Set We consider optimization over a convex set $\mathcal{K} \subseteq [0, 1]^d$. We assume $\mathcal{K}$ is *down-closed*, meaning that for any $\mathbf{y} \in \mathcal{K}$ and any $\mathbf{0} \leq \mathbf{x} \leq \mathbf{y}$, we have $\mathbf{x} \in \mathcal{K}$. Additionally, we assume that:

Assumption 1 (Geometry of Constraint Set). The constraint set $\mathcal{K}$ has bounded diameter $D > 0$, i.e., $\max_{\mathbf{x}, \mathbf{y} \in \mathcal{K}} \|\mathbf{x} - \mathbf{y}\| \leq D$.

To ensure computational efficiency, we assume access to a base online linear optimization algorithm $\mathcal{A}$ that is efficient for the constraint set $\mathcal{K}$ using either projection-based or projection-free (e.g., Frank-Wolfe, SO-OGA) updates.

DR-Submodularity We focus on non-negative, differentiable functions $f : [0, 1]^d \rightarrow \mathbb{R}_{\geq 0}$. A function f over $\mathcal{K}$ is called *continuous DR-submodular* if $\forall \mathbf{x} \leq \mathbf{y} \in \mathcal{K}$, we have $\nabla f(\mathbf{x}) \geq \nabla f(\mathbf{y})$. Equivalently, if f is twice-differentiable, all entries of its Hessian matrix are non-positive ($\nabla^2 f(\mathbf{x}) \leq \mathbf{0}$). We explicitly *do not* assume monotonicity; entries of $\nabla f(\mathbf{x})$ may be negative. Additionally,

we assume that:

Assumption 2 (Regularity of Objective Function). The functions $f_t \in \mathcal{F}$ are M_1 -Lipschitz continuous, i.e., $\|\nabla f_t(\mathbf{x})\| \leq M_1$ for all $\mathbf{x} \in \mathcal{K}$, and L -smooth, i.e., $\|\nabla f_t(\mathbf{x}) - \nabla f_t(\mathbf{y})\| \leq L\|\mathbf{x} - \mathbf{y}\|$.

3.2. Problem Setting: Adversarial Online Optimization

We consider a standard adversarial online optimization of time horizon T with function class $\mathcal{F}$ over constraint set $\mathcal{K}$. At round t , the player selects a pair of points $\hat{\mathbf{x}}_t, \mathbf{u}_t \in \mathcal{K}$, an adversary reveals a objective function $f_t \in \mathcal{F}$ with its query oracle $\mathbf{O}_t$. Then the player plays $\hat{\mathbf{x}}_t$, queries $\mathbf{O}_t$ at $\mathbf{u}_t$, and receives feedback $\mathbf{o}_t$ and updates its decision.

The query oracle is the only way for the player to learn about the functions selected by the adversary. In our work, for the main algorithm, we assume that:

Assumption 3 (Unbiased Gradient Oracle). For the functions f_t , the algorithm has access to a stochastic gradient oracle $\mathbf{O}$ such that for any query point $\mathbf{x}_t$, it returns a vector $\mathbf{g}_t$ satisfying that:

$$\mathbb{E}[\mathbf{g}_t \mid \mathbf{x}_t] = \nabla f_t(\mathbf{x}_t) \quad \text{and} \quad \|\mathbf{g}_t\| \leq B_1.$$

3.3. Limited Feedback Setting

When $\mathbf{u}_t = \hat{\mathbf{x}}_t$, we say the queries are trivial, as the points of query are the same as the played actions; otherwise, we say the queries are non-trivial. We say the provided oracle $\mathbf{O}_t$ is first-order if it returns the gradient of the given function at the point of query, or zeroth-order if it returns the value of the function.

When the player have non-trivial queries, we say the player takes full-information feedback, which can be either first-order or zeroth-order. When the player has trivial queries, we say the player takes semi-bandit feedback if the adversarial provide first-order oracles, or bandit feedback if zeroth-order oracles are provided.

In Section 5, we begin our analysis with algorithm under first-order full-information feedback, and gives regret analysis. Then we consider zeroth-order full-information feedback, semi-bandit feedback, and bandit feedback,

3.4. Regret

Our goal is to minimize the α -regret, which measures the gap between a algorithm and an α -approximate static optimum. α is often referred to as the optimal approximation ratio. The α -static regret is defined as:

$$\mathcal{R}_\alpha(T) \triangleq \alpha \max_{\mathbf{x} \in \mathcal{K}} \sum_{t=1}^T f_t(\mathbf{x}) - \sum_{t=1}^T \mathbb{E}[f_t(\mathbf{x}_t)]. \quad (1)$$

Beyond static regret, we also consider robustness to non-stationary environments via two advanced metrics. **Adaptive regret** is defined as the maximum regret over any contiguous interval $[s, e] \subseteq [T]$, i.e.,

$$\mathcal{AR}_\alpha(T) \triangleq \sup_{[s, e] \subseteq [T]} \left\{ \alpha \max_{\mathbf{x} \in \mathcal{K}} \sum_{t=s}^e f_t(\mathbf{x}) - \sum_{t=s}^e \mathbb{E}[f_t(\mathbf{x}_t)] \right\}.$$

Dynamic regret is defined as the regret against a sequence of time-varying comparators $\mathbf{u}_t^* \triangleq \arg \max_{\mathbf{u} \in \mathcal{K}} f_t(\mathbf{u})$, bounded by the path length $P_T = \sum_{t=1}^T \|\mathbf{u}_t^* - \mathbf{u}_{t-1}^*\|$, i.e.,

$$\mathcal{DR}_\alpha(T) \triangleq \alpha \sum_{t=1}^T f_t(\mathbf{u}_t^*) - \sum_{t=1}^T \mathbb{E}[f_t(\mathbf{x}_t)]. \quad (2)$$

Remark 1 (Optimal Approximation Ratio α). Maximizing non-monotone DR-submodular functions is NP-hard, even in the offline setting. Formally, this means the best efficient algorithm can only guarantee a value of α times the optimal value; thus, α is called the optimal approximation ratio. For monotone objectives, algorithms can achieve a stronger approximation ratio of $1 - 1/e$ (if the constraint set contains the origin) or $1/2$ (for general convex sets). However, these monotonicity-based benchmarks are fundamentally unachievable in the non-monotone regime. For non-monotone functions, no polynomial-time algorithm can achieve an approximation ratio better than 0.478 (Gharan & Vondrák, 2011; Qi, 2024). While recent offline algorithms have narrowed this gap by achieving a 0.401-approximation (Buchbinder & Feldman, 2024), the best known guarantee for efficient online algorithms remains $1/e \approx 0.367$ (Thang & Srivastav, 2021; Zhang et al., 2023).

3.5. Linearizability

A core tool in our analysis is the *linearizable* framework introduced by Pedramfar & Aggarwal (2024b). This property allows us to reduce non-convex DR-submodular optimization to online linear optimization.

Definition 1 (Linearizability). A function f over $\mathcal{K}$ is α -linearizable if there exists a mapping $h : \mathcal{K} \rightarrow \mathcal{K}$ and a vector field $\mathbf{g} : \mathcal{F} \times \mathcal{K} \rightarrow \mathbb{R}^d$ such that for all $\mathbf{x}, \mathbf{y} \in \mathcal{K}$:

$$\alpha f(\mathbf{y}) - f(h(\mathbf{x})) \leq \beta \langle \mathbf{g}(f, \mathbf{x}), \mathbf{y} - \mathbf{x} \rangle. \quad (3)$$

Intuitively, the definition reduces online non-convex DR-submodular maximization to online linear optimization (OLO): it upper-bounds the scaled reward $\alpha f(\mathbf{y})$ (up to $f(h(\mathbf{x}))$) by the linear term $\langle \mathbf{g}(f, \mathbf{x}), \mathbf{y} - \mathbf{x} \rangle$. Thus, any low-regret OLO algorithm run on $\mathbf{g}(f_t, \mathbf{x}_t)$ transfers to α -regret for the original rewards, up to the constant β (and the mapping h). We call h reparameterization, β scaling parameter, and $\mathbf{g}$ the surrogate potential. Here α is exactly the approximation/competitive factor in our regret benchmark.

4. Non-monotone DR-submodular Functions over Down-Closed Convex Sets are $1/e$ -Linearizable

As previously described, in this paper, we consider the case where the constraint set $\mathcal{K} \subseteq [0, 1]^d$ is a down-closed convex set containing the origin. We show that the class of non-monotone DR-submodular functions over such sets is linearizable with $1/e$ -approximation ratio.

A central challenge for proving upper-linearizability of non-monotone functions over $\mathcal{K}$ is to construct a global lower bound that relates the algorithm's trajectory to an arbitrary comparator $\mathbf{y}$ (e.g., the global optimum). Unlike monotone regimes where simple greedy algorithms suffice, the non-monotone landscape requires balance between ascending gradients and shrinking constraints.

The following key Lemma overcomes this by establishing a novel structural inequality for our exponential reparameterization $h_z(\mathbf{x})$. This inequality is a mathematical specialization of Lemma 4.1 from Buchbinder & Feldman (2024) to the constant-path case. By leveraging the down-closed property of the constraint set, this formulation provides the exact structural piece required to accommodate the adversarial online setting. It proves that this specific reparameterization preserves a strict lower bound relative to any target $\mathbf{y}$, effectively bridging the gap between the DR-submodular property and our $1/e$ approximation ratio.

Lemma 1. *Let f be a non-monotone continuous DR-submodular function over a down-closed convex set. For any $\mathbf{x}, \mathbf{y} \in [0, 1]^d$ and $z \in [0, 1]$, let $h_z(\mathbf{x}) = \mathbf{1} - e^{-z\mathbf{x}}$, we have:*

$$f(h_z(\mathbf{x}) \oplus \mathbf{y}) \geq e^{-z\bar{x}} f(\mathbf{y}) \geq e^{-z} f(\mathbf{y}) \quad (4)$$

where $\bar{x} = \max_j [\mathbf{x}]_j$. (here $[\mathbf{x}]_j$ is the j -th component of the vector $\mathbf{x}$).

The proof of Lemma 1 is provided in Appendix B. Next, we have the main result of this paper:

Theorem 1 (Main Theorem). *Let $\mathcal{K} \subseteq [0, 1]^d$ be a down-closed convex set such that $\mathbf{0} \in \mathcal{K}$. Let $f : [0, 1]^d \rightarrow \mathbb{R}_{\geq 0}$ be a non-negative, differentiable, non-monotone DR-submodular function. Define the mapping $h : \mathcal{K} \rightarrow \mathcal{K}$ as $h(\mathbf{x}) \triangleq \mathbf{1} - e^{-\mathbf{x}}$ and $h_z(\mathbf{x}) \triangleq \mathbf{1} - e^{-z\mathbf{x}}$. Let the vector field $\mathbf{g} : \mathcal{F} \times \mathcal{K} \rightarrow \mathbb{R}^d$ be $\mathbf{g}(f, \mathbf{x}) \triangleq \nabla F(\mathbf{x})$, where $F : \mathcal{K} \rightarrow \mathbb{R}$ is the function defined by:*

$$F(\mathbf{x}) \triangleq \int_0^1 \frac{e^{z-1}}{(1-e^{-1})z} (f(\mathbf{1} - e^{-z\mathbf{x}}) - f(\mathbf{0})) dz. \quad (5)$$

Then for $\forall \mathbf{x}, \mathbf{y} \in \mathcal{K}$, the following inequality holds:

$$\frac{1}{e} f(\mathbf{y}) - f(h(\mathbf{x})) \leq (1 - e^{-1}) \langle \mathbf{g}(f, \mathbf{x}), \mathbf{y} - \mathbf{x} \rangle. \quad (6)$$

Thus, the function f is α -linearizable with approximation coefficient $\alpha = \frac{1}{e}$, the scaling parameter $\beta = 1 - e^{-1}$, the reparameterization $h(\mathbf{x}) = \mathbf{1} - e^{-\mathbf{x}}$, and surrogate potential $\mathbf{g}$.

Remark 2. Intuitively, the exponential mapping $h(\mathbf{x})$ is specifically chosen because it ensures that the reparameterized trajectory strictly remains within the valid domain for downward-closed convex sets. In the same time, $F(\mathbf{x})$ acts as an integrating factor that heavily weights the early stages of the trajectory, which enables an exact mathematical cancellation of the non-monotone penalty during the integration-by-parts step of the subsequent analysis.

Proof. Clearly we have $F(\mathbf{0}) = 0$. For any $\mathbf{x} \neq \mathbf{0}$, the integrand in Equation (5) is a continuous function of z that is bounded by:

$$\begin{aligned} & \frac{e^{z-1}}{(1-e^{-1})z} (f(\mathbf{1} - e^{-z\mathbf{x}}) - f(\mathbf{0})) \\ & \stackrel{(a)}{\leq} \frac{e^{z-1}}{(1-e^{-1})z} M_1 \|\mathbf{1} - e^{-z\mathbf{x}}\| \stackrel{(b)}{\leq} \frac{1}{1-e^{-1}} M_1, \end{aligned}$$

where (a) follows from the M_1 -Lipschitz continuity of f , and (b) uses the bound $\|\mathbf{1} - e^{-z\mathbf{x}}\| \leq 1$. Therefore F is well-defined on $[0, 1]^d$.

Next, differentiating F with respect to $\mathbf{x}$,

$$\begin{aligned} \nabla F(\mathbf{x}) &= \int_0^1 \frac{e^{z-1}}{(1-e^{-1})z} \nabla (f(h_z(\mathbf{x})) - f(\mathbf{0})) dz \\ & \stackrel{(a)}{=} \int_0^1 \frac{e^{z-1}}{(1-e^{-1})z} \left(\nabla f(h_z(\mathbf{x})) \odot \frac{\partial}{\partial \mathbf{x}} h_z(\mathbf{x}) \right) dz \\ & \stackrel{(b)}{=} \int_0^1 \frac{e^{z-1}}{1-e^{-1}} (\nabla f(h_z(\mathbf{x})) \odot e^{-z\mathbf{x}}) dz. \quad (7) \end{aligned}$$

where (a) uses the chain rule w.r.t. $\mathbf{x}$, and (b) uses $\frac{\partial}{\partial \mathbf{x}} h_z(\mathbf{x}) = z e^{-z\mathbf{x}}$ because $h_z(\mathbf{x}) = \mathbf{1} - e^{-z\mathbf{x}}$.

Now, we use (7), and $\mathbf{x}$ is independent of z , we have

$$\begin{aligned} & (1 - e^{-1}) \langle \nabla F(\mathbf{x}), -\mathbf{x} \rangle \\ & \stackrel{(a)}{=} \int_0^1 e^{z-1} \langle \nabla f(h_z(\mathbf{x})) \odot e^{-z\mathbf{x}}, -\mathbf{x} \rangle dz \\ & \stackrel{(b)}{=} \int_0^1 e^{z-1} \langle \nabla f(h_z(\mathbf{x})), -\mathbf{x} \odot e^{-z\mathbf{x}} \rangle dz, \\ & \stackrel{(c)}{=} \int_0^1 e^{z-1} \left(-\frac{d}{dz} f(h_z(\mathbf{x})) \right) dz \quad (8) \end{aligned}$$

where (a) uses linearity of the inner product and interchange of integral and inner product, (b) uses the identity $\langle \mathbf{a} \odot \mathbf{b}, \mathbf{c} \rangle = \langle \mathbf{a}, \mathbf{b} \odot \mathbf{c} \rangle$, and (c) uses the chain rule in z that $\frac{d}{dz} f(h_z(\mathbf{x})) = \langle \nabla f(h_z(\mathbf{x})), \frac{d}{dz} h_z(\mathbf{x}) \rangle = \langle \nabla f(h_z(\mathbf{x})), \mathbf{x} \odot e^{-z\mathbf{x}} \rangle$.

We now apply integration by parts to (8) with $u(z) = e^{z-1}$ and $dv(z) = -\frac{d}{dz}f(h_z(\mathbf{x}))dz$. Then $du(z) = e^{z-1}dz$ and $v(z) = -f(h_z(\mathbf{x}))$. Hence,

$$\begin{aligned} (1 - e^{-1})\langle \nabla F(\mathbf{x}), -\mathbf{x} \rangle &= \left[-e^{z-1}f(h_z(\mathbf{x})) \right]_0^1 + \int_0^1 e^{z-1}f(h_z(\mathbf{x}))dz \\ &= -f(h_1(\mathbf{x})) + \frac{1}{e}f(h_0(\mathbf{x})) + \int_0^1 e^{z-1}f(h_z(\mathbf{x}))dz \\ &\geq -f(h(\mathbf{x})) + \int_0^1 e^{z-1}f(h_z(\mathbf{x}))dz, \end{aligned}$$

since $h_1(\mathbf{x}) = h(\mathbf{x})$ and $h_0(\mathbf{x}) = \mathbf{0}$.

Again, using (7), we have

$$\begin{aligned} (1 - e^{-1})\langle \nabla F(\mathbf{x}), \mathbf{y} \rangle &= \int_0^1 e^{z-1}\langle \nabla f(h_z(\mathbf{x})) \odot e^{-z\mathbf{x}}, \mathbf{y} \rangle dz \\ &\stackrel{(a)}{=} \int_0^1 e^{z-1}\langle \nabla f(h_z(\mathbf{x})), \mathbf{y} \odot e^{-z\mathbf{x}} \rangle dz \\ &\stackrel{(b)}{=} \int_0^1 e^{z-1}\langle \nabla f(h_z(\mathbf{x})), \mathbf{y} \odot (1 - h_z(\mathbf{x})) \rangle dz \\ &\stackrel{(c)}{\geq} \int_0^1 e^{z-1}[f(h_z(\mathbf{x}) \oplus \mathbf{y}) - f(h_z(\mathbf{x}))] dz \end{aligned}$$

where (a) uses the identity $\langle \mathbf{a} \odot \mathbf{b}, \mathbf{c} \rangle = \langle \mathbf{a}, \mathbf{b} \odot \mathbf{c} \rangle$, (b) is due to $h_z(\mathbf{x}) = \mathbf{1} - e^{-z\mathbf{x}}$, and (c) uses Lemma 3 and the fact that $h_z(\mathbf{x}) \oplus \mathbf{y} = \mathbf{1} - (\mathbf{1} - h_z(\mathbf{x})) \odot (\mathbf{1} - \mathbf{y}) = h_z(\mathbf{x}) + \mathbf{y} \odot (1 - h_z(\mathbf{x}))$.

Lemma 1 states that, $\forall \mathbf{x}, \mathbf{y} \in [0, 1]^d$ and $z \in [0, 1]$, we have $f(h_z(\mathbf{x}) \oplus \mathbf{y}) \geq e^{-z}f(\mathbf{y})$. Thus,

$$\begin{aligned} (1 - e^{-1})\langle \nabla \mathcal{F}(\mathbf{x}), \mathbf{y} \rangle &\geq \int_0^1 e^{z-1}[e^{-z}f(\mathbf{y}) - f(h_z(\mathbf{x}))] dz \\ &\geq f(\mathbf{y}) \int_0^1 e^{z-1}e^{-z} dz - \int_0^1 e^{z-1}f(h_z(\mathbf{x})) dz \\ &= f(\mathbf{y}) \int_0^1 e^{-1} dz - \int_0^1 e^{z-1}f(h_z(\mathbf{x})) dz \\ &= \frac{1}{e}f(\mathbf{y}) - \int_0^1 e^{z-1}f(h_z(\mathbf{x})) dz. \end{aligned} \tag{9}$$

Summing both terms:

$$\begin{aligned} (1 - e^{-1})\langle \mathbf{g}(f, \mathbf{x}), \mathbf{y} - \mathbf{x} \rangle &= (1 - e^{-1})\langle \nabla F(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \\ &= (1 - e^{-1})[\langle \nabla \mathcal{F}(\mathbf{x}), \mathbf{y} \rangle + \langle \nabla \mathcal{F}(\mathbf{x}), -\mathbf{x} \rangle] \\ &= \frac{1}{e}f(\mathbf{y}) - f(h(\mathbf{x})) \end{aligned}$$

because the integral terms $\int_0^1 e^{z-1}f(h_z(\mathbf{x}))dz$ cancels. $\square$

Remark 3 (Expectation form of ∇F). Let $\mathcal{Z} \in [0, 1]$ be a random variable with CDF

$$\mathbb{P}(\mathcal{Z} \leq z) = \int_0^z \frac{e^{u-1}}{1 - e^{-1}} du, \tag{10}$$

equivalently with density $p(z) = \frac{e^{z-1}}{1 - e^{-1}}$ on $[0, 1]$. Then (7) admits the expectation representation

$$\nabla F(\mathbf{x}) = \mathbb{E}_{\mathcal{Z}}[\nabla f(h_{\mathcal{Z}}(\mathbf{x})) \odot e^{-\mathcal{Z}\mathbf{x}}]. \tag{11}$$

In order to attain an unbiased and bounded estimate of $\mathbf{g}$, we provide Boosted Query algorithm for Non-monotone DR-submodular functions over Down-closed convex set (BQND), detailed in Algorithm 1, and we formally state these properties in Lemma 2. Algorithm 1 is necessary to provide regret guarantee for our main algorithm, as it satisfies the conditions of Theorem 3, the Regret Transfer Theorem. Theorem 3 establishes the fundamental regret transfer guarantee for our framework, proving that the difficult problem of non-monotone maximization strictly reduces to the simpler problem of online linear optimization. It formalizes the linearizable reduction, guaranteeing that the regret of our main algorithm is bounded by the regret of the linear base learner on the surrogate functions, up to a scaling constant β . This allows us to directly inherit the convergence rates of the chosen base solver.

Algorithm 1 Boosted Query algorithm for Non-monotone DR-Submodular functions over Down-closed convex set (BQND)

- 1: **Input:** Point $\mathbf{x} \in \mathcal{K}$, first-order stochastic oracle $\mathbf{O}$ for f .
- 2: **Sampling:** Sample $z \in [0, 1]$ according to Equation 10
- 3: **Query:** Construct point of query $\mathbf{u} = \mathbf{1} - e^{-z\mathbf{x}}$.
- 4: Query the first-order oracle $\mathbf{O}$ which returns a stochastic sample of $\nabla f(\mathbf{u})$, denoted as $\mathbf{v}$.
- 5: **Output:** Return the estimator $\mathbf{g} = \mathbf{v} \odot e^{-z\mathbf{x}}$.

Lemma 2 (Properties of BQND Estimator). *Let $\mathbf{g}$ be the output of Algorithm 1 for an input $\mathbf{x} \in \mathcal{K}$ and a first-order oracle for function f that satisfies Assumption 3. Then $\mathbf{g}$ satisfies that:*

$$\mathbb{E}[\mathbf{g}] = \mathbf{g}(f, \mathbf{x}) = \nabla F(\mathbf{x}), \text{ and } \|\mathbf{g}\| \leq B_1.$$

i.e., Algorithm 1 returns an unbiased estimate of $\mathbf{g}$ that is bounded by B_1 .

Proof. From Algorithm 1, the output is $\mathbf{g} = \mathbf{v} \odot e^{-z\mathbf{x}}$, where under Assumption 3, $\mathbb{E}[\mathbf{v}] = \nabla f(\mathbf{1} - e^{-z\mathbf{x}})$ and z is

sampled according to Equation 10. Taking the expectation over z :

$$\mathbb{E}[\mathbf{g}] = \int_0^1 \frac{e^{z-1}}{1-e^{-1}} (e^{-z\mathbf{x}} \odot \nabla f(\mathbf{1} - e^{-z\mathbf{x}})) dz.$$

This integral is identical to the gradient derivation of the surrogate function $F(\mathbf{x})$ proved in Theorem 1. Thus, $\mathbb{E}[\mathbf{g}] = \nabla F(\mathbf{x}) = \mathbf{g}(\mathbf{x})$.

The norm of the output is:

$$\|\mathbf{g}\| = \|\mathbf{v} \odot e^{-z\mathbf{x}}\| \leq \|\mathbf{v}\| \cdot \max_i |e^{-zx_i}|$$

Since $\mathbf{x} \in \mathcal{K} \subseteq [0, 1]^d$ and $z \geq 0$, the term $0 < e^{-zx_i} \leq 1$ for all i . Therefore, the element-wise shrinking cannot increase the norm. Using Assumption 3, we have

$$\|\mathbf{g}\| \leq B_1 \cdot 1 = B_1. \quad \square$$

5. Regret Results Applied from Linearizable Optimization

A key advantage of the linearizable formulation is that it allows us to seamlessly transfer guarantees from Online Linear/Convex Optimization to our non-monotone DR-submodular setting. To begin with, we consider the most common setting, where the adversary provides a first-order noisy oracle as described in Assumption 3. We propose Algorithm 2, a modular reduction framework for maximizing non-monotone DR-submodular functions over down-closed convex sets.

The algorithm requires initializing two subroutines: a base learner $\mathcal{A}$, which can be any efficient regret-minimizing algorithm for online linear optimization over $\mathcal{K}$, and a query algorithm $\mathcal{G}$ that is fixed to be BQND (Algorithm 1). While our framework is compatible with any such linear solver, to obtain the specific regret guarantees in this paper, we instantiate $\mathcal{A}$ with Online Gradient Ascent via Separation Oracle (SO-OGA) (Appendix C) or Improved Ader (IA) (Appendix E). The base learner receives first-order feedback and updates its decision to $\mathbf{x}_t$. Our main algorithm then plays the action $\hat{\mathbf{x}}_t$ and passes $\mathbf{x}_t$ to the query algorithm, which interacts with the oracle to return an unbiased estimate of $\mathbf{g}$.

Algorithm 2 follows the reduction-to-linear-optimization paradigm for linearizable functions, instantiating the Online Maximization By Quadraticization (OMBQ) meta-algorithm proposed by Pedramfar & Aggarwal (2024b) (see Appendix D.1) with our specific exponential mapping $h(\mathbf{x}) = \mathbf{1} - e^{-\mathbf{x}}$ and the BQND query algorithm. Because Theorem 1 successfully establishes the structural

¹Specific instantiations of $\mathcal{A}$ for our results are given in the propositions.

Algorithm 2 Adaptive Projection-free Online Non-monotone DR-Submodular Maximization over Down-closed Convex Sets

- 1: **Input:** Horizon T , Constraint set $\mathcal{K}$, Stepsize η .
- 2: **Initialize:**
- 3: Base Learner $\mathcal{A}$: Any online linear optimization algorithm with sublinear regret.¹
- 4: Query Algorithm $\mathcal{G}$: BQND (Algorithm 1).
- 5: **for** $t = 1, \dots, T$ **do**
- 6: Receive action $\mathbf{x}_t$ from Base Learner $\mathcal{A}$
- 7: Play action $\hat{\mathbf{x}}_t = \mathbf{1} - e^{-\mathbf{x}_t}$
- 8: Adversary selects a function f_t and a first-order query oracle $\mathbf{O}_t$.
- 9: Run query algorithm: $\mathbf{o}_t \leftarrow \mathcal{G}(\mathbf{O}_t, \mathbf{x}_t)$.
- 10: Pass $\mathbf{o}_t$ to $\mathcal{A}$ to update its state.
- 11: **end for**

$1/e$ -linearizability of our domain, the subsequent regret guarantees across various feedback models do not require independent non-convex analyses. Instead, the bounds presented in the following subsections and summarized in Table 1 follow naturally as direct mechanical corollaries of plugging Theorem 1 into the OMBQ framework alongside appropriate linear base learners. This configuration yields the first algorithm for this domain with improved regret permitting $\mathcal{O}(1)$ oracle queries per round.

5.1. Static Regret

Our framework (Algorithm 2) works with any base learner $\mathcal{A}$ that minimizes regret for linear functions. To derive improved regret rate that is efficient, we instantiate $\mathcal{A}$ with the projection-free learner SO-OGA in Appendix C. This algorithm is proposed by Garber & Kretzu (2022) and later refined by Pedramfar & Aggarwal (2024b). The result is as follows:

Proposition 1 (Static Regret). *Let $\{f_t\}_{t=1}^T$ be a sequence of M_1 -Lipschitz non-monotone DR-submodular functions. Assume we have access to unbiased estimates of the gradient that are bounded by $B_1 \geq M_1$. Instantiating base learner to be SO-OGA as described in Algorithm 3, Algorithm 2 achieves a $1/e$ -approximation static regret bounded by:*

$$\mathbb{E} \left[\alpha \max_{\mathbf{u} \in \mathcal{K}} \sum_{t=1}^T f_t(\mathbf{u}) - \sum_{t=1}^T f_t(\hat{\mathbf{x}}_t) \right] \leq \mathcal{O}(B_1 \sqrt{T}) \quad (12)$$

for any fixed comparator $\mathbf{y} \in \mathcal{K}$. The algorithm requires only $\mathcal{O}(1)$ gradient queries per round.

Proof. Since we choose the base learner to be SO-OGA, it follows from Theorem 2 that

$$\mathcal{R}_1^{\text{SO-OGA}} = \mathcal{O}(B_1 T^{1/2})$$

In Lemma 2, we proved that Algorithm 1 returns an unbiased bounded estimate of $\mathbf{g}$. Thus, using Theorem 3, we have:

$$\mathbb{E} \left[\alpha \max_{\mathbf{u} \in \mathcal{K}} \sum_{t=1}^T f_t(\mathbf{u}) - \sum_{t=1}^T f_t(\hat{\mathbf{x}}_t) \right] \leq \beta \mathcal{R}_1^{\text{SO-OGA}} = \mathcal{O}(M_1 T^{1/2}) \quad \square$$

5.2. Adaptive and Dynamic Regrets

Since SO-OGA is known to minimize adaptive regret for linear functions (Garber & Kretzu, 2022; Pedramfar & Aggarwal, 2024b), Algorithm 2 instance in Proposition 1 automatically achieves an adaptive regret bound described below:

Proposition 2 (Adaptive Regret). *Let $\{f_t\}_{t=1}^T$ be a sequence of M_1 -Lipschitz non-monotone DR-submodular functions over down-closed convex sets. Assume we have access to unbiased estimates of the gradient that are bounded by $B_1 \geq M_1$. Instantiating base learner to be SO-OGA as described in Algorithm 3, Algorithm 2 achieves a $1/e$ -approximation adaptive regret bounded by:*

$$\mathcal{AR}_{1/e}(T) \leq \mathcal{O}(T^{1/2}) \quad (13)$$

This is the first adaptive regret guarantee for non-monotone DR-submodular maximization over down-closed sets.

Proof. If we choose the base learner to be SO-OGA, it follows from Theorem 2 that

$$\mathcal{AR}_1^{\text{SO-OGA}} = \mathcal{O}(B_1 T^{1/2})$$

In Lemma 2, we proved that Algorithm 1 returns an unbiased bounded estimate of $\mathbf{g}$. Thus, using Theorem 3, we have:

$$\mathcal{AR}_{1/e}(T) \leq \beta \mathcal{R}_1^{\text{SO-OGA}} = \mathcal{O}(T^{1/2}). \quad \square$$

To handle non-stationary environments, we instantiate Algorithm 2 with IMPROVED ADER algorithm (Algorithm 6) in Appendix E as the base linear learner $\mathcal{A}$. This algorithm is proposed by Zhang et al. (2018) and later refined by Pedramfar & Aggarwal (2024b). The result is as follows:

Proposition 3 (Dynamic Regret Guarantees). *Let $P_T = \sum_{t=1}^T \|\mathbf{x}_t^* - \mathbf{x}_{t-1}^*\|$ denote the path length of the sequence of optimal minimizers for the surrogate linear functions. Let $\{f_t\}_{t=1}^T$ be a sequence of M_1 -Lipschitz non-monotone DR-submodular functions over down-closed convex sets. Assume we have access to unbiased estimates of the gradient that are bounded by $B_1 \geq M_1$. Instantiating base learner to be IA as described in Algorithm 6, Algorithm 2 achieves a $1/e$ -approximation dynamic regret bounded by:*

$$\mathcal{DR}_{1/e}(T) \leq \tilde{\mathcal{O}} \left(\sqrt{T(1 + P_T)} \right). \quad (14)$$

Proof. If we choose the base learner to be IA for linear functions, it follows from Theorem 4 that

$$\mathcal{R}_{1,L}^{\text{IA}}(\mathbf{u}) = \mathcal{O} \left(B_1 \sqrt{T(1 + P_T(\mathbf{u}))} \right)$$

In Lemma 2, we proved that Algorithm 1 returns an unbiased bounded estimate of $\mathbf{g}$. Thus, using Theorem 3, we have:

$$\mathcal{DR}_{1/e}(\mathbf{u}) \leq \beta \mathcal{R}_1^{\text{IA}}(\mathbf{u}) = \tilde{\mathcal{O}}(\sqrt{T(1 + P_T)}) \quad \square$$

5.3. Semi-Bandit, Zeroth-order Full-Information, and Bandit Feedback

Recall that for Algorithm 2, we assumed first-order gradient, and the queries determined by the query algorithm BQND are non-trivial. Thus, the results we obtained in Theorem 1, 2, and 3 are given for first-order full-information feedback. By decoupling the non-convex query algorithm from the linear base learner, our modular design allows seamless extensions to restrictive feedback settings. In this section, we show that by applying meta-algorithms developed for linearizable functions by Pedramfar & Aggarwal (2024b), our regret guarantees naturally transfer to Semi-Bandit, Zeroth-Order Full-Information, and Bandit settings. Furthermore, we provide the first theoretical analysis extending these results to Adaptive and Dynamic regret measures, establishing a unified projection-free framework that is robust to both limited information and non-stationary feedback.

Given first-order oracles, when only trivial queries are permitted, we say that the algorithm handles semi-bandit feedback. Thus, we apply SFTT (Appendix D.3), with $\mathcal{A}^{\text{action}}$ being the SO-OGA and $\mathcal{A}^{\text{query}}$ being BQND. Thus, applying Lemma 7 with $\eta = 1/2$ due to Proposition 1 and 2, we have the following result:

Proposition 4 (Semi-Bandit Guarantees). *Let $\{f_t\}_{t=1}^T$ be a sequence of non-monotone DR-submodular functions over down-closed convex sets. If we instantiate SFTT meta-algorithm with block size $L = T^{1/3}$, using SO-OGA as the base learner $\mathcal{A}^{\text{action}}$ and BQND as the query algorithm $\mathcal{A}^{\text{query}}$, the resulting algorithm achieves the following regret bounds against $\{f_t\}_{t=1}^T$:*

$$\mathbb{E}[\mathcal{R}(T)] \leq \mathcal{O}(T^{2/3}) \quad \text{and} \quad \mathcal{AR}_{1/e}(T) \leq \mathcal{O}(T^{2/3}).$$

When provided only zeroth-order oracles (i.e., oracles returning value estimates for functions instead of gradient estimates), we say that the algorithm handles zeroth-order full-information feedback if the queries are non-trivial, or bandit feedback if the queries are trivial. For zeroth-order full-information feedback, we apply FOTZO (Appendix D.2), with $\mathcal{A}$ being the SO-OGA and $\mathcal{A}^{\text{query}}$ being BQND. Thus, applying Lemma 5 with $\eta = 1/2$ due to Proposition 1 and 2, we have the following result:

Proposition 5 (Zeroth-Order Full-Information Guarantees). *Let $\{f_t\}_{t=1}^T$ be a sequence of non-monotone DR-submodular functions over down-closed convex sets. Let Algorithm $\mathcal{A}$ be the instantiation of the FOTZO using BQND as $\mathcal{A}^{query}$ equipped with base algorithm SO-OGA. By employing a one-point gradient estimator with smoothing radius δ , the resulting Algorithm $\mathcal{A}$ achieves the following regret bounds against $\{f_t\}_{t=1}^T$:*

$$\mathbb{E}[\mathcal{R}(T)] \leq O(T^{3/4}) \quad \text{and} \quad \mathcal{AR}_{1/e}(T) \leq O(T^{3/4}).$$

If only a zeroth-order oracle is provided, and only trivial queries are allowed, we say that the algorithm handles bandit feedback. For such limited feedback, we apply STB meta-algorithm (Appendix D.2) to the algorithm instantiated in Proposition 4, and applying Lemma 6, we obtain the following results:

Proposition 6 (Bandit Guarantees). *Let $\{f_t\}_{t=1}^T$ be a sequence of non-monotone DR-submodular functions over down-closed convex sets. Let Algorithm $\mathcal{A}$ be the semi-bandit algorithm instantiated in Proposition 4. Applying STB, the resulting Algorithm achieves the following regret bounds against $\{f_t\}_{t=1}^T$:*

$$\mathbb{E}[\mathcal{R}(T)] \leq O(T^{2/3}) \quad \text{and} \quad \mathcal{AR}_{1/e}(T) \leq O(T^{2/3}).$$

Note that the reduction lemmas (Lemmas 7, 5, and 6) transfer the regret guarantees from first-order full-information setting to limited settings regardless of the choice of base learner. Consequently, by instantiating the meta-algorithms with the IMPROVED ADER base learner (Proposition 3), we obtain dynamic regret bounds for all limited feedback settings.

Proposition 7 (Dynamic Regret for Limited Feedback). *The expected dynamic regret $\mathbb{E}[\mathcal{R}_{1/e}^{dyn}(T)]$ for the described algorithm in the corresponding proposition is bounded by $\tilde{O}(T^{2/3}\sqrt{1+P_T})$ for Semi-Bandit feedback, $\tilde{O}(T^{3/4}\sqrt{1+P_T})$ for Zeroth-Order Full-Info feedback, and $\tilde{O}(T^{5/6}\sqrt{1+P_T})$ for Bandit feedback. where P_T is the path length of the optimal sequence.*

6. Conclusion and Future Work

Conclusions. In this work, we resolved a critical theoretical bottleneck in online non-monotone DR-submodular maximization over down-closed convex sets by proving that the function class is strictly $1/e$ -linearizable. By carefully designing an exponential reparameterization and a surrogate potential that yields an exact cancellation during integration-by-parts, we successfully decoupled the $1/e$ approximation limit from the computationally heavy Frank-Wolfe mechanics. Coupled with our BQND gradient estimator, this structural breakthrough enabled a clean reduction to standard Online Linear Optimization utilizing only

a strict single-query budget per round. As a result, we were able to elevate the state of the art across multiple feedback models, providing $O(T^{1/2})$ static regret alongside the first adaptive and dynamic regret guarantees for this domain.

Limitations and Future Work. While our linearizable reduction achieves the best-known online approximation ratio $1/e$ with $O(T^{1/2})$ regret, a theoretical gap remains compared to the best achievable offline approximation of 0.401 (Buchbinder & Feldman, 2024). A fundamental limitation in the online setting is the adversarial revelation of the non-monotone penalty, which currently prevents algorithms from making the globally aware corrective steps utilized by state-of-the-art offline methods. A direction for future work is determining whether this $1/e$ online barrier can be broken, perhaps by exploring predictive feedback models or relaxed constraint geometries. Additionally, because the optimal online approximation ratio remains an open question, establishing strict, end-to-end regret lower bounds for the limited-feedback regimes explored in this paper represents another crucial open challenge. Finally, extending the results to γ -weakly DR-submodular functions, as studied in (Pedramfar et al., 2024b; Jadav et al., 2026), remains an open direction.

Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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A. Useful Lemmas

For non-negative, continuous, non-monotone DR-submodular functions over down-closed convex sets, (Buchbinder & Feldman, 2024) established several inequalities that we use in our analysis.

Lemma 3 (Lemma 2.1, Buchbinder & Feldman (2024)). *Let f be a non-negative continuous DR-submodular function over $[0, 1]^d$. Then, $\forall \mathbf{x} \in [0, 1]^d$ and $\mathbf{y} \geq \mathbf{0}$ such that $\mathbf{x} + \mathbf{y} \leq \mathbf{1}$, we have:*

$$\langle \nabla f(\mathbf{x}), \mathbf{y} \rangle \geq f(\mathbf{x} + \mathbf{y}) - f(\mathbf{x}).$$

Lemma 4 (Lemma 4.1, Buchbinder & Feldman (2024)). *Let $f : [0, 1]^d \rightarrow \mathbb{R}_{\geq 0}$ be a non-negative DR-submodular function. Given $t \geq 0$, an integrable function $\mathbf{x} : [0, t] \rightarrow [0, 1]^d$, and a vector $\mathbf{a} \in [0, 1]^d$, the original lemma states:*

$$f\left(\mathbf{1} - \mathbf{a} \odot e^{-\int_0^t \mathbf{x}(\tau) d\tau}\right) \geq e^{-t} \cdot \left[f(\mathbf{1} - \mathbf{a}) + \sum_{i=1}^{\infty} \frac{1}{i!} \cdot \int_{\tau \in [0, t]^i} f\left((\mathbf{1} - \mathbf{a}) \oplus \bigoplus_{j=1}^i \mathbf{x}(\tau_j)\right) d\tau \right] \quad (15)$$

B. Proof of Lemma 1

Proof. In (15), for online adversarial setting, we assume $\mathbf{x}(\tau) = \mathbf{x}$ is constant over the interval $[0, t]$, so on the left hand side, $\int_0^t \mathbf{x}(\tau) d\tau = t\mathbf{x}$, and on the right hand side, $\bigoplus_{j=1}^i \mathbf{x}(\tau_j) = \mathbf{1} - \bigodot_{j=1}^i (\mathbf{1} - \mathbf{x}) = \mathbf{1} - (\mathbf{1} - \mathbf{x})^i$, and because the function F takes values in $\mathbb{R}_{\geq 0}$ (it is non-negative), every term inside the integral is non-negative. Consequently, the entire infinite sum is non-negative:

$$\begin{aligned} \sum_{i=1}^{\infty} \frac{1}{i!} \cdot \int_{\tau \in [0, t]^i} f\left((\mathbf{1} - \mathbf{a}) \oplus \bigoplus_{j=1}^i \mathbf{x}(\tau_j)\right) d\tau &= \sum_{i=1}^{\infty} \frac{1}{i!} \cdot \int_{\tau \in [0, t]^i} f\left((\mathbf{1} - \mathbf{a}) \oplus (\mathbf{1} - (\mathbf{1} - \mathbf{x})^i)\right) d\tau \\ &= \sum_{i=1}^{\infty} \frac{1}{i!} \cdot \int_{\tau \in [0, t]^i} f\left(\mathbf{1} - \mathbf{a} \odot (\mathbf{1} - \mathbf{x})^i\right) d\tau \\ &= \sum_{i=1}^{\infty} \frac{t^i}{i!} \cdot f\left(\mathbf{1} - \mathbf{a} \odot (\mathbf{1} - \mathbf{x})^i\right) \geq 0 \end{aligned}$$

By dropping the non-negative sum term from the RHS, we obtain a strictly weaker, but simpler lower bound:

$$f\left(\mathbf{1} - \mathbf{a} \odot e^{-z\mathbf{x}}\right) \geq e^{-t} \cdot f(\mathbf{1} - \mathbf{a})$$

Let $\mathbf{y} = \mathbf{1} - \mathbf{a} \in [0, 1]^d$. Let the time parameter t be z . Then,

$$f\left(\mathbf{1} - (\mathbf{1} - \mathbf{y}) \odot e^{-z\mathbf{x}}\right) \geq e^{-z} f(\mathbf{y})$$

Let $h_z(\mathbf{x}) = \mathbf{1} - e^{-z\mathbf{x}}$. By definition of probabilist sum $\oplus$, $h_z(\mathbf{x}) \oplus \mathbf{y} := \mathbf{1} - (\mathbf{1} - \mathbf{y}) \odot e^{-z\mathbf{x}}$. Thus,

$$f(h_z(\mathbf{x}) \oplus \mathbf{y}) \geq e^{-z} f(\mathbf{y})$$

Define $\bar{x} \triangleq \max_j [\mathbf{x}]_j$ where $[\mathbf{x}]_j$ is the j -th component of the vector $\mathbf{x}$. Thus, $\bar{x} \in [0, 1]$. Thus, we have

$$f(h_z(\mathbf{x}) \oplus \mathbf{y}) \geq e^{-z\bar{x}} f(\mathbf{y}) \geq e^{-z} f(\mathbf{y})$$

□

C. Infeasible Projection and Online Gradient Ascent via a Separation Oracle (SO-OGA)

To bypass the computational bottleneck of Euclidean projections in high-dimensional spaces (e.g., $\mathcal{O}(n^3)$ SVD for trace-norm balls), Frank-Wolfe type projection-free methods using Linear Optimization Oracles (LOO) have become standard, following the foundational work of Hazan & Kale (2012) and Jaggi (2013). However, standard LOO-based methods often face suboptimal convergence rates in adversarial online settings. Addressing this, Garber & Kretzu (2022) introduced Separation Oracle (SO) based methods, which can achieve $\mathcal{O}(T^{1/2})$ regret. To efficiently solve our surrogate linear optimization problem, we utilize their SO-OGA algorithm (adapted by Pedramfar & Aggarwal (2024b)).

Remark 4 (Oracles for Constraint Set - LOO and SO). To circumvent the high cost of Euclidean projections, we rely on two natural projection-free oracles. Given a convex set $\mathcal{K}$ and a query point $\mathbf{y}$, the Linear Optimization Oracle (LOO) returns $\arg \min_{\mathbf{x} \in \mathcal{K}} \langle \mathbf{y}, \mathbf{x} \rangle$, while the Separation Oracle (SO) either asserts $\mathbf{y} \in \mathcal{K}$ or returns a separating hyperplane $\mathbf{g}$ such that $\forall \mathbf{x} \in \mathcal{K}, \langle \mathbf{g}, \mathbf{y} - \mathbf{x} \rangle > 0$. While LOO is more prevalent, these oracles are *complementary*. As noted by Garber & Kretzu (2022), for the nuclear norm ball, the LOO is efficient while the SO is expensive. Conversely, for the *spectral norm ball* $\mathcal{B}_2$, the situation is reversed. Thus, the SO enables efficient online learning over domains where the LOO is computationally intractable.

We utilize the SO-OGA instantiation and its subroutine SO-IP from Pedramfar & Aggarwal (2024b). We adopt the notation from the original paper: let $\mathbf{c} \in \text{int}(\mathcal{K})$ be a center point, $r > 0$ be the radius of a ball contained in $\mathcal{K}$ centered at $\mathbf{c}$, and define the shrunk set $\hat{\mathcal{K}}_\delta \triangleq \{(1 - \delta/r)(\mathbf{x} - \mathbf{c}) + \mathbf{c} \mid \mathbf{x} \in \mathcal{K}\}$.

Algorithm 3 Online Gradient Ascent via Separation Oracle - SO-OGA (Algorithm 8 in Pedramfar & Aggarwal (2024b))

- 1: **Input:** horizon T , constraint set $\mathcal{K}$, step size η .
 - 2: **Initialize:** $\mathbf{x}_1 \leftarrow \mathbf{c} \in \hat{\mathcal{K}}_\delta$.
 - 3: **for** $t = 1, 2, \dots, T$ **do**
 - 4: Play $\mathbf{x}_t$ and observe $\mathbf{o}_t = \nabla f_t(\mathbf{x}_t)$.
 - 5: $\mathbf{x}'_{t+1} = \mathbf{x}_t + \eta \mathbf{o}_t$ {Gradient Ascent Step}
 - 6: Set $\mathbf{x}_{t+1} = \text{SO-IP}_{\mathcal{K}}(\mathbf{x}'_{t+1})$. {Output of Algorithm 4}
 - 7: **end for**
-

Algorithm 4 Infeasible Projection via Separation Oracle - SO-IP $_{\mathcal{K}}(\mathbf{y}_0)$ (Algorithm 9 in Pedramfar & Aggarwal (2024b))

- 1: **Input:** Constraint set $\mathcal{K}$, shrinking parameter $\delta < r$, initial point $\mathbf{y}_0$.
 - 2: $\mathbf{y}_1 \leftarrow \mathbf{P}_{\text{aff}(\mathcal{K})}(\mathbf{y}_0)$
 - 3: $\mathbf{y}_2 \leftarrow \mathbf{c} + \frac{\mathbf{y}_1 - \mathbf{c}}{\max\{1, \|\mathbf{y}_1\|/D\}}$ {Projection of $\mathbf{y}_0$ over $\mathbb{B}_D^d(\mathbf{c}) \cap \text{aff}(\mathcal{K})$ }
 - 4: **for** $i = 1, 2, \dots$ **do**
 - 5: Call Separation Oracle $\text{SO}_{\mathcal{K}}$ with input $\mathbf{y}_i$.
 - 6: **if** $\mathbf{y}_i \notin \mathcal{K}$ **then**
 - 7: Set $\mathbf{g}_i$ to be the hyperplane returned by $\text{SO}_{\mathcal{K}}$ (i.e., $\forall \mathbf{x} \in \mathcal{K}, \langle \mathbf{y}_i - \mathbf{x}, \mathbf{g}_i \rangle > 0$).
 - 8: $\mathbf{g}'_i \leftarrow \mathbf{P}_{\text{aff}(\mathcal{K}) - \mathbf{c}}(\mathbf{g}_i)$
 - 9: Update $\mathbf{y}_{i+1} \leftarrow \mathbf{y}_i - \delta \frac{\mathbf{g}'_i}{\|\mathbf{g}'_i\|}$.
 - 10: **else**
 - 11: **Return** $\mathbf{y} \leftarrow \mathbf{y}_i$.
 - 12: **end if**
 - 13: **end for**
-

Theorem 2 (Adaptive Regret of SO-OGA, Theorem 9 in Pedramfar & Aggarwal (2024b)). *Let $\mathbf{L}$ be a class of linear functions over $\mathcal{K}$ such that $\|l\| \leq M_1$ for all $l \in \mathbf{L}$ and let $D = \text{diam}(\mathcal{K})$. Fix $v > 0$ such that $\delta = vT^{-1/2} \in (0, 1)$ and set the step size $\eta = \frac{vr}{2M_1}T^{-1/2}$. Then we have:*

$$\mathcal{AR}_{1, \text{Adv}_1^f(\mathbf{L})}^{\text{SO-OGA}} = O(M_1 T^{1/2}). \quad (16)$$

D. Meta-Algorithms and Regret Bounds Towards Diverse Feedback Types

D.1. The Generic Meta-Algorithm (OMBQ)

We rely on the generic reduction framework established in Pedramfar & Aggarwal (2024b). The following meta-algorithm, OMBQ, transforms a linear learner $\mathcal{A}$ into a non-monotone DR-submodular maximizer using a mapping h and a query oracle $\mathcal{G}$.

Theorem 3 (Regret Transfer, Theorem 1 in Pedramfar & Aggarwal (2024b)). *Let $\mathcal{A}$ be an algorithm for online optimization with semi-bandit feedback. Also let $\mathcal{F}$ be a function class over $\mathcal{K}$ that is linearizable and surrogate potential $\mathbf{g} : \mathcal{F} \times \mathcal{K} \rightarrow \mathbb{R}^d$ and $h : \mathcal{K} \rightarrow \mathcal{K}$. Let $\mathcal{G}$ be a query algorithm for $\mathbf{g}$ and let $\mathcal{A}' = \text{OMBQ}(\mathcal{A}, \mathcal{G}, h)$.*

Algorithm 5 Online Maximization By Quadratization - OMBQ($\mathcal{A}, \mathcal{G}, h$)

- 1: **Input:** Base algorithm $\mathcal{A}$, Query algorithm $\mathcal{G}$, Mapping $h : \mathcal{K} \rightarrow \mathcal{K}$.
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: Let $\mathbf{x}_t$ be the action chosen by $\mathcal{A}$.
- 4: **Play:** $\mathbf{y}_t = h(\mathbf{x}_t)$.
- 5: **Query:** Call oracle $\mathcal{G}$ at $\mathbf{x}_t$ to obtain gradient estimate $\mathbf{g}_t$.
- 6: **Update:** Pass loss vector $-\mathbf{g}_t$ to $\mathcal{A}$ to update its state.
- 7: **end for**

If $\mathcal{G}$ returns an unbiased estimate of $\mathbf{g}$ and the output of $\mathcal{G}$ is bounded by B_1 , then we have:

$$\mathcal{R}_{\alpha, \text{Adv}_1^{\mathcal{Q}}(\mathbf{F}, B_1)}^{\mathcal{A}'} \leq \beta \mathcal{R}_{1, \text{Adv}_1^f(\mathbf{Q}_\mu[B_1])}^{\mathcal{A}} \quad (17)$$

where β is a scaling constant.

For the following meta-algorithms including FOTZO, STB, and SFTT, we refer the exact implementation to the updated manuscripts by [Pedramfar & Aggarwal \(2024a\)](#) hosted on Arxiv.

D.2. First Order To Zeroth Order (FOTZO) and Semi-bandit To Bandit (STB)

Let $T_\delta(\mathbf{x}) = (1 - \frac{\delta}{r})\mathbf{x} + \frac{\delta}{r}\mathbf{c}$ and $T^\delta(\mathbf{x}) = \frac{\mathbf{x} - \frac{\delta}{r}\mathbf{c}}{1 - \frac{\delta}{r}}$. Let $f : T_\delta([0, 1]^d) \rightarrow \mathbb{R}_{\geq 0}$ be a non-negative, differentiable, non-monotone DR-submodular function. Let $\mathcal{K} \subseteq T_\delta([0, 1]^d)$ be a convex set such that $\mathcal{K}^\delta := T^\delta(\mathcal{K})$ is downward-closed. Define $f^\delta := f \circ T_\delta : [0, 1]^d \rightarrow \mathbb{R}_{\geq 0}$. Let $\mathbf{F}'$ denote the function class f^δ over $\mathcal{K}^\delta$.

Lemma 5 (Corollary 4 + Theorem 5 in [Pedramfar & Aggarwal \(2024a\)](#)). *Let $\mathbf{F}$ be an M_1 -Lipschitz function class over a convex set $\mathcal{K}$ and choose $\mathbf{c}$ and r as described above and let $\delta < \frac{r}{2}$. Let $\mathcal{U} \subseteq \mathcal{K}^T$ be a compact set. Assume $\mathcal{A}$ is an algorithm for online optimization with first order feedback. Then, if $\mathcal{A}' = \text{FOTZO}(\mathcal{A})$ and $0 < \alpha \leq 1$, we have*

$$\mathcal{R}_{\alpha, \text{Adv}_0^{\mathcal{Q}}(\mathbf{F}, B_0)}^{\mathcal{A}'}(\mathcal{U}) \leq \mathcal{R}_{\alpha, \text{Adv}_1^{\mathcal{Q}}(\mathbf{F}', \frac{2k}{\delta}B_0)}^{\mathcal{A}}(\hat{\mathcal{U}}) + \left(3 + \frac{2D}{r}\right) \delta M_1 T.$$

Further, if we have $\mathcal{R}_{\alpha, \text{Adv}_1^{\mathcal{Q}}(\mathbf{F}', B_1)}^{\mathcal{A}} = O(B_1 T^\eta)$ and $\delta = T^{(\eta-1)/2}$, then we have

$$\mathcal{R}_{\alpha, \text{Adv}_0^{\mathcal{Q}}(\mathbf{F}, B_0)}^{\mathcal{A}'} = O(B_0 T^{(1+\eta)/2}).$$

Lemma 6 (Corollary 5 + Theorem 6 in [Pedramfar & Aggarwal \(2024a\)](#)). *Under the assumptions of Lemma 5, if we assume that $\mathcal{A}$ is semi-bandit, then the same regret bounds hold with $\mathcal{A}' = \text{STB}(\mathcal{A})$, where STB is described by Algorithm 6. Further, if we have $\delta = T^{-1}$, then $\mathcal{R}_{\alpha, \text{Adv}_0^{\mathcal{Q}}(\mathbf{F})}^{\mathcal{A}'}$ has the same order of regret as that of $\mathcal{R}_{\alpha, \text{Adv}_1^{\mathcal{Q}}(\mathbf{F}', B_1)}^{\mathcal{A}}$ with B_1 replaced with kM_1 .*

D.3. Stochastic Full-information To Trivial query (SFTT)

Lemma 7 (Corollary 6 + Theorem 7 in [Pedramfar & Aggarwal \(2024a\)](#)). *Let $\mathcal{A}$ be an online optimization algorithm with full-information feedback and with K queries at each time-step where $\mathcal{A}^{\text{query}}$ does not depend on the observations in the current round and $\mathcal{A}' = \text{SFTT}(\mathcal{A})$. Then, for any M_1 -Lipschitz function class $\mathbf{F}$ that is closed under convex combination and any $B_1 \geq M_1$, $0 < \alpha \leq 1$ and $1 \leq a \leq b \leq T$, let $a' = \lfloor (a-1)/L \rfloor + 1$, $b' = \lceil b/L \rceil$, $D = \text{diam}(\mathcal{K})$ and let $\{T\}$ and $\{T/L\}$ denote the horizon of the adversary. Then, we have*

$$\mathcal{R}_{\alpha, \text{Adv}_i^{\mathcal{Q}}(\mathbf{F}, B_1)\{T\}}^{\mathcal{A}'}(\mathcal{K}_*^T)[a, b] \leq M_1 DK(b' - a' + 1) + L \mathcal{R}_{\alpha, \text{Adv}_1^{\mathcal{Q}}(\mathbf{F}, B_1)\{T/L\}}^{\mathcal{A}}(\mathcal{K}_*^{T/L})[a', b'],$$

Further, if we have $\mathcal{R}_{\alpha, \text{Adv}_i^{\mathcal{Q}}(\mathbf{F}, B)}^{\mathcal{A}}(\mathcal{K}_*^T)[a, b] = O(BT^\eta)$, $K = O(T^\theta)$ and $L = T^{\frac{1+\theta-\eta}{2-\eta}}$, then we have

$$\mathcal{R}_{\alpha, \text{Adv}_i^{\mathcal{Q}}(\mathbf{F}, B)}^{\mathcal{A}'}(\mathcal{K}_*^T)[a, b] = O\left(BT^{\frac{(1+\theta)(1-\eta)+\eta}{2-\eta}}\right).$$

As a special case, when $K = O(1)$, then we have

$$\mathcal{R}_{\alpha, \text{Adv}_i^{\mathcal{Q}}(\mathbf{F}, B)}^{\mathcal{A}'}(\mathcal{K}_*^T)[a, b] = O\left(BT^{\frac{1}{2-\eta}}\right).$$

E. Algorithms for Dynamic Regret

To support the dynamic regret guarantees presented in Proposition 3, we restate the IMPROVED ADER meta-algorithm and its corresponding expert algorithm from Pedramfar & Aggarwal (2024b). We also include the main theorem governing its performance.

Algorithm 6 Improved Ader - IA (Restated Algorithm 10 from Pedramfar & Aggarwal (2024b))

- 1: **Input:** horizon T , constraint set $\mathcal{K}$, step size λ , a set $\mathcal{H}$ containing step sizes for experts.
 - 2: Activate a set of experts $\{E^\eta \mid \eta \in \mathcal{H}\}$ by invoking Algorithm 7 for each step size $\eta \in \mathcal{H}$.
 - 3: Sort step sizes in ascending order $\eta_1 \leq \dots \leq \eta_N$, and set $w_1^{\eta_i} = \frac{C}{i(i+1)}$ where $C = 1 + \frac{1}{|\mathcal{H}|}$.
 - 4: **for** $t = 1, 2, \dots, T$ **do**
 - 5: Receive $\mathbf{x}_t^\eta$ from each expert E^η .
 - 6: Play the action $\mathbf{x}_t = \sum_{\eta \in \mathcal{H}} w_t^\eta \mathbf{x}_t^\eta$ and observe $\mathbf{o}_t = \nabla f_t(\mathbf{x}_t)$.
 - 7: Define $\ell_t(\mathbf{y}) := \langle \mathbf{o}_t, \mathbf{y} - \mathbf{x}_t \rangle$.
 - 8: Update the weight of each expert by $w_{t+1}^\eta = \frac{w_t^\eta e^{-\lambda \ell_t(\mathbf{x}_t^\eta)}}{\sum_{\mu \in \mathcal{H}} w_t^\mu e^{-\lambda \ell_t(\mathbf{x}_t^\mu)}}$.
 - 9: Send the gradient $\mathbf{o}_t$ to each expert E^η .
 - 10: **end for**
-

Algorithm 7 Improved Ader : Expert algorithm (Restated Algorithm 11 from (Pedramfar & Aggarwal, 2024b))

- 1: **Input:** horizon T , constraint set $\mathcal{K}$, step size η .
 - 2: Let $\mathbf{x}_1^\eta$ be any point in $\mathcal{K}$.
 - 3: **for** $t = 1, 2, \dots, T$ **do**
 - 4: Send $\mathbf{x}_t^\eta$ to the main algorithm.
 - 5: Receive $\mathbf{o}_t$ from the main algorithm.
 - 6: **Update:** $\mathbf{x}_{t+1}^\eta = \mathbf{P}_{\mathcal{K}}(\mathbf{x}_t^\eta + \eta \mathbf{o}_t)$
 - 7: *{Note: To maintain the projection-free property of our framework, we implement this update using the Frank-Wolfe step or the Infeasible Projection subroutine from Algorithm 4.}*
 - 8: **end for**
-

Theorem 4 (Dynamic Regret of Improved Ader, Theorem 10 in Pedramfar & Aggarwal (2024b)). *Let $\mathbf{L}$ be a class of linear functions over $\mathcal{K}$ such that $\|\ell\| \leq M_1$ for all $\ell \in \mathbf{L}$ and let $D = \text{diam}(\mathcal{K})$. Set $\mathcal{H} := \{\eta_i = \frac{2^{i-1}D}{M_1} \sqrt{\frac{7}{2T}} \mid 1 \leq i \leq N\}$ where $N = \lceil \frac{1}{2} \log_2(1 + 4T/7) \rceil + 1$ and $\lambda = \sqrt{2/(TM_1^2 D^2)}$. Then for any comparator sequence $\mathbf{u} \in \mathcal{K}^T$, we have*

$$\mathcal{R}_{1, \text{Adv}_1^f(\mathbf{L})}^{\text{IA}}(\mathbf{u}) = O\left(M_1 \sqrt{T(1 + P_T(\mathbf{u}))}\right) \quad (18)$$

where $P_T(\mathbf{u}) = \sum_{t=1}^T \|\mathbf{u}_t - \mathbf{u}_{t-1}\|_2$ is the path length of the comparator sequence.